

VISHNU GAMINI

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EDUCATION

University of Southern California

M.S., Computer Science (GPA: 3.82/4.0)

• **Coursework:** Machine Learning, Deep Learning, Applied Natural Language Processing, Analysis of Algorithms, Agentic AI

Aug 2025 - Present

Los Angeles, CA

Vellore Institute of Technology

B.Tech, Computer Science (GPA: 9.12/10)

Jun 2021 - Jul 2025

Amaravati, India

SKILLS

Core Competencies: Software Engineering, Machine Learning, Backend Systems

Languages: Python, Java, C++, HTML, CSS, JavaScript, SQL, TypeScript

ML/AI: PyTorch, TensorFlow, OpenCV, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), LangChain, Deep Learning, Generative AI, Scikit-learn, Pandas, NumPy

Frameworks & Tools: FastAPI, Flask, ROS, Node.js, Docker, Kubernetes, Git, Claude Code, Microservices, CI/CD, React, Cursor, Codex

Cloud: AWS, GCP, Azure, Oracle Cloud

Databases: MongoDB, MySQL, PostgreSQL, Vector Databases

EXPERIENCE

University of Southern California | Graduate Research Assistant (Interaction Lab)

Apr 2026 - Present

- Engineered automated data collection and preprocessing pipelines using custom web scrapers and Python scripts to extract, clean, and convert over 50,000 psychotherapy dialogue samples into structured, training-ready formats on a Linux-based on-premise CUDA-enabled GPU server managed via SSH.
- Fine-tuned Llama 3.1 on psychotherapy dialogue data using LoRA/PEFT to generate more contextually aligned and conversationally entrained therapeutic responses, and integrated the model with the Blossom robotic platform through ROS for real-time inference.
- Fine-tuned a BERT-based conversational entrainment model in PyTorch using self-supervised contrastive learning, optimizing dialogue embeddings so entrained conversation pairs achieved higher cosine similarity than negative pairs and improving the mean cosine-similarity margin by 18% over the baseline.

VersaQuant | Software Development Engineer Intern

Mar 2025 - May 2025

- Developed an automated warehouse safety monitoring system in Python served via FastAPI, using OpenCV to preprocess surveillance footage frames fed through a transfer-learned VGG16 and ResNet50 ensemble to detect, classify, and timestamp safety violations. Dockerized the PyTorch-backed FastAPI inference service, with confidence-scored outputs logged to DynamoDB and flagged video clips stored in S3 for structured downstream review, enabling faster identification and audit of safety incidents.
- Built React and TypeScript monitoring dashboards to track predictions, confidence scores, and model drift, helping maintain 92% accuracy on unseen data.

eAppSys | Machine Learning Intern

Aug 2023 - Jan 2024

- Architected and deployed an agentic RAG conversational AI system on OCI using LangChain and Pinecone, implementing a ReAct-based reasoning loop with dynamic tool invocation across APIs, databases, and web search, and exposing the system through a FastAPI backend that improved client query resolution efficiency by 15%.
- Built an automated data ingestion and indexing pipeline using Scrapy and Apache Airflow, orchestrating scheduled web crawls, content preprocessing, Hugging Face sentence-transformer embedding generation, and vector upserts into Pinecone; enabled sub-second semantic retrieval across the company's knowledge base.

VIT-AP University | Undergraduate Research Assistant

Mar 2023 - Oct 2024

- Improved skin lesion segmentation Dice score by 12% by integrating SE channel recalibration and CBAM spatial attention into a U-Net architecture, enhancing focus on lesion boundary regions; used NumPy for image tensor preprocessing, with CUDA acceleration and an OpenCV augmentation pipeline incorporating rotation, flipping, and elastic deformation for improved generalization on imbalanced medical imaging data.
- Built end-to-end TensorFlow/Keras training pipelines with mixed-precision training and Bayesian hyperparameter optimization, evaluated on Dice and IoU metrics; used pandas to organize experiment results, compare model performance, and support metric analysis; co-authored research on attention-based medical image segmentation accepted to Multimedia Tools and Applications.

PROJECTS

Local LLM Agent | [GitHub](#)

- Built a fully autonomous Python-based LLM agent with a custom agentic harness and reflection-based self-correction loop that analyzes runtime outputs, identifies failures, and autonomously rewrites and retries until completion, supporting end-to-end workflows across code generation, debugging, file system management, and GitHub integration from a single natural language prompt.
- Engineered a custom multi-agent orchestration engine with conditional routing, inter-agent delegation, and domain-specialized sub-agents spanning coding, error correction, search, and file management, safe on-demand code execution through a secondary LLM review layer, and a Flask full-stack interface that organically amassed 20+ GitHub stars validating real-world utility.

Archaic Style Conversation Generator | [GitHub](#)

- Built a 5M-parameter decoder-only Transformer from scratch in PyTorch with Grouped Query Attention (GQA), Rotary Positional Embeddings (RoPE), and a BPE tokenizer; trained with a custom autoregressive loop with KV-cache inference optimization for archaic-style dialogue generation.
- Converted a trained PyTorch model to ONNX Runtime and applied INT8 quantization for efficient edge inference, deploying it as a Dockerized FastAPI service on a Raspberry Pi running Linux.

AI Interview Coach | [GitHub](#)

- Architected an AI interview preparation platform using Next.js, TypeScript, and Supabase (authentication, PostgreSQL database, and file storage), integrated with OpenAI APIs for resume/JD ingestion, PDF/DOCX parsing, role-context extraction, and company-aware coding round generation.
- Engineered a multi-agent pipeline (planner, research, question-generation, and evaluator agents) that dynamically sequences interview rounds and draws questions from real candidate interview reports through live web research and candidate repositories through GitHub MCP, delivered through an adaptive Monaco-based code editor with protected API routes, autosaved drafts, and persistent round history.

Secure One Time Chat | [GitHub](#)

- Built a real-time chat app with Node.js and WebSockets, with password-protected ephemeral rooms and zero server-side message persistence.
- Implemented a GitHub Actions CI/CD pipeline to run automated tests, validate builds, and deploy the full-stack application to Vercel.